2026-10-02 | VS&A; DAILY | 10-MINUTE LESSON

Four feedlines, one sump: coherence decides the combined tone

Learning objective. Decide when multiple tonal pressure contributions should be added as complex amplitudes, combined as root-sum-square power, or represented by a cross-spectral matrix.

Suppose four pumps generate a tone near the same order. Each feedline attenuates and phase-shifts its source before the disturbances enter a common sump. Asking for the pressure from “all four lines” is incomplete: the answer depends on whether the arriving signals retain a stable phase relationship over the observation interval.

The two familiar limits differ sharply. Four equal, mutually incoherent sources produce twice one-source pressure RMS, a +6.02 dB power increase. Four equal, perfectly in-phase tones produce four times the pressure RMS, +12.04 dB. Coherent cancellation can drive the result below either value. RSS is therefore neither automatically conservative nor automatically nonconservative.

Physical separation does not guarantee incoherence. Sources can be phase-locked by a common shaft, control system, combustion coupling or shared acoustic network. Conversely, nominally identical rotating machines can drift enough that their relative phase walks through many cycles during an analysis record. Coherence is therefore a property of the signals, paths, operating condition and observation interval together. The appropriate model can change between startup, a steady plateau and a transient sweep even when the individual tone levels appear similar.

Coherent tone: $\tilde{p}_{sum}(f) = \Sigma\, H_i(f)\, \tilde{s}_i(f)$
Random process: $G_{pp}(f) = h^H(f)\, S_{ss}(f)\, h(f)$

The first equation adds complex RMS phasors. The second is its stochastic counterpart: S_{ss} contains every source autospectrum and cross-spectrum, while h contains the complex path transfers. Discarding the off-diagonal terms asserts incoherence. That is a modeling choice, not a neutral simplification.

All geometry, pressures, phases and calculations in this lesson are illustrative. They are not measured data from a launch vehicle or a prediction for a particular cryogenic system.

01 / PRESERVE COMPLEX TRANSFER

A loss model without phase is incomplete

A one-dimensional line model propagates pressure and volume-flow perturbations through distributed segments and lumped elements. NASA CR-124308 describes feedline models that calculate both terminal-pressure amplitude and phase while including effects such as mean flow, compliances, bubbles, bellows and structural-hydraulic coupling [1].

A minimal traveling-wave placeholder for path i is $H_i = |H_i| \exp[-j\, kL_i]$. It is useful for sensitivity studies, but reflections, impedance changes, side branches, screens and a finite receiving volume generally require a coupled network solution. Fluid-network acoustic modes can be formulated from pressure and flow-rate transmission-line equations [2].

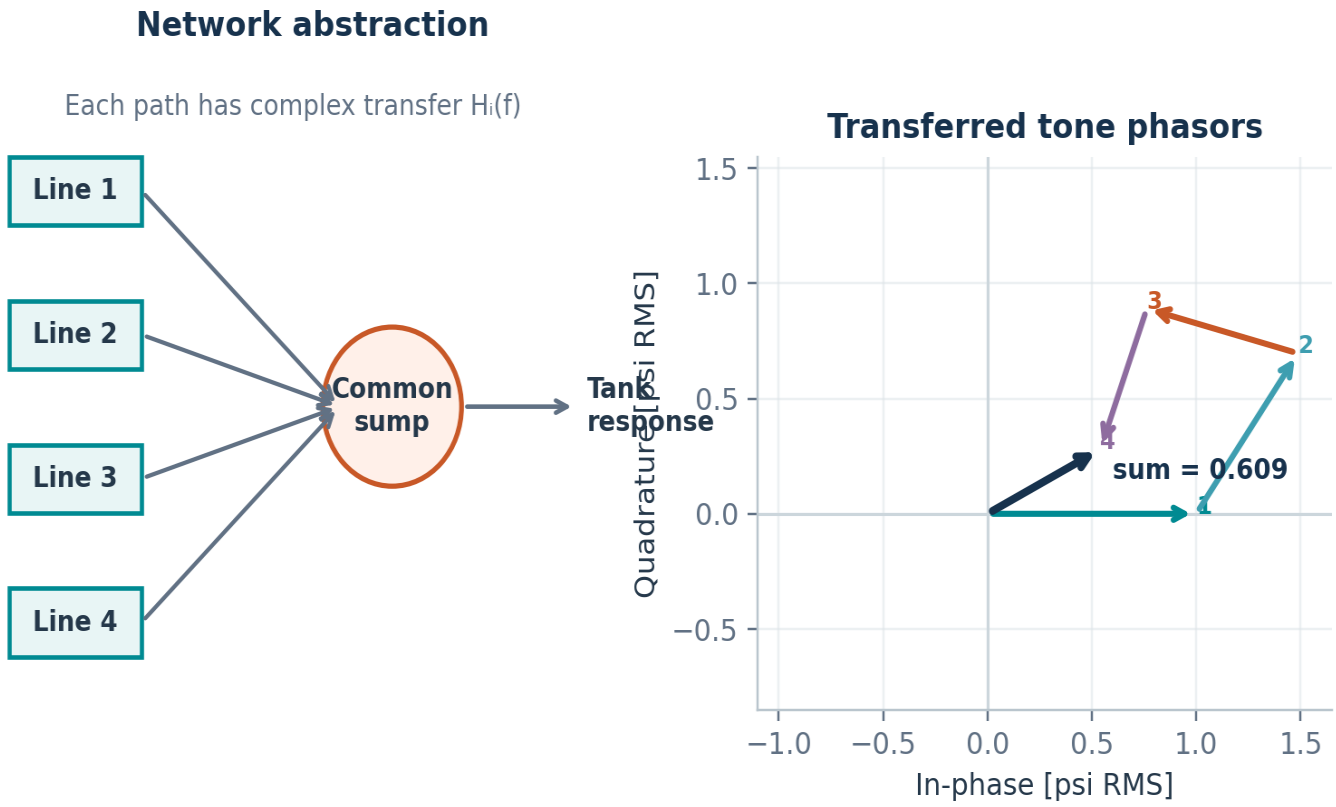


Figure 1. Original schematic and phasor construction. The four arrows on the right are the transmitted RMS pressure contributions at one common plane, not the source pressures. Their vector sum is shown in navy.

The sump or tank is not necessarily a one-way receiver. Its input impedance changes each branch reflection coefficient, and parallel branches interact through their shared boundary. Solve the assembled network when that feedback matters; adding four independently terminated line solutions can violate the actual junction conditions.

Once pressure reaches the tank, wall response adds another complex transfer. A single pressure measurement or a single panel acceleration may sit near a node. Use spatially distributed pressure and structural response predictions when the objective is tank-wall load or stress.

02 / THE CROSS TERMS CARRY THE ANSWER

Coherence magnitude is not phase

For deterministic tones at one frequency, let $q_i = H_i \tilde{s}_i$ be the RMS phasor arriving from line i. The combined mean-square pressure is

$$p_{rms,sum}^2 = |\Sigma q_i|^2 = \Sigma |q_i|^2 + 2\,\mathrm{Re}\{\Sigma_{i<j}\, q_i q_j^*\}$$

RSS keeps only the first sum. The cross terms can be positive or negative. For stationary random processes the same information appears in complex cross-spectra G_{ij}. The magnitude-squared coherence

$$\gamma_{ij}^2(f) = |G_{ij}(f)|^2 / [G_{ii}(f) G_{jj}(f)]$$

ranges from zero to one, but it does not carry the sign of constructive or destructive interference. You also need the cross-spectrum phase. NISTIR 4733 defines cross-spectral density as a generally complex quantity and discusses phase and coherence estimation [3]. NASA engine-noise work likewise used correlation and coherence to recover relative amplitude and phase between internal and radiated pressures [4].

A high coherence estimate can mean a stable linear relationship, but low coherence is not proof of independent physics. Unrelated noise, multiple unmeasured inputs, nonlinearity, frequency drift, time variation and insufficient averaging can all reduce the estimate. Inspect time-frequency behavior before assigning an incoherent model.

For N channels, S_{ss} must be Hermitian and positive semidefinite. Pairwise coherence magnitudes chosen independently may violate that constraint and yield negative predicted power. Build the matrix from simultaneous data, a consistent source model, or a valid factorization.

03 / WORKED EXAMPLE

The same amplitudes support three answers

Assume the transmitted tone contributions at the sump are 1.00, 0.85, 0.75 and 0.65 psi RMS with phases 0, 55, 165 and 250 degrees. These are invented post-transfer values. They deliberately include unequal loss and phase.

Combination rule	Result [psi RMS]	Level relative to 1 psi
Arithmetic sum; all in phase	3.250	+10.24 dB
RSS; mutually incoherent	1.645	+4.33 dB
Complex sum; stated phases	0.609	-4.31 dB
Physical phasor bound	0 to 3.250	depends on relative phase

The stated phases produce 0.609 psi RMS: 8.64 dB below the RSS estimate. If all four phases align, the same magnitudes produce 3.25 psi RMS: 5.91 dB above RSS. Neither extreme changes the individual line amplitudes.

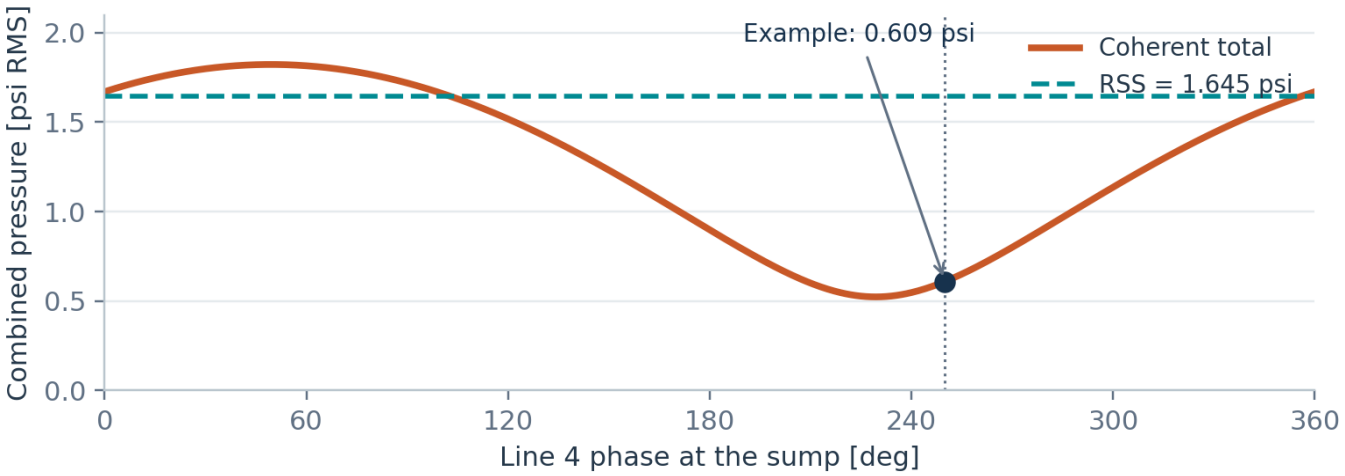


Figure 2. Original sensitivity calculation. The first three phasors stay fixed while only line 4 phase is swept. The combined pressure ranges from 0.523 to 1.823 psi RMS; the dashed RSS value is not a bound.

Phase can be highly sensitive to path length. Using an illustrative wave speed c = 1000 m/s, not an LH2 property, a 0.10 m length change at 1800 Hz produces 64.8 degrees of traveling-wave phase. Effective wave speed may also change with wall compliance, void fraction, temperature and mean flow. A nominal cancellation should therefore be treated as fragile unless geometry and medium state are controlled.

04 / WHEN A TONE IS NOT ONE TONE

Frequency drift changes the combining rule

Two sources can share an order without sharing an exact frequency. If their frequencies differ by Δf, their relative phase rotates at 2πΔf. The measured pressure beats, and a finite-time RMS includes a cross term whose average depends on ΔfT. For a rectangular observation of duration T, the normalized cross-term magnitude follows a sinc-like factor.

$$\text{cross-term retention} \propto \operatorname{sinc}(\pi\,\Delta f\,T)$$

If |Δf|T is much greater than one, the cross term largely averages away and power addition becomes reasonable for that record. If |Δf|T is small, the tones remain effectively coherent. Between those limits, the reported answer depends on record length, window, resolution bandwidth and whether order tracking follows each source.

Measurement plan. Acquire all line pressures and the sump or tank pressure simultaneously with common timing. Preserve complex spectra. Examine order-tracked phase, ordinary and partial coherence, and segment-to-segment stability. Use enough spectral resolution to separate close tones; otherwise two drifting lines can occupy one bin and create misleading coherence.

Model plan. Propagate each source through a complex line model, assemble junction and receiving impedances, and carry a source CSD matrix. Evaluate at least three cases: measured/estimated coherence, incoherent RSS, and a physically admissible coherent upper envelope. Include phase uncertainty from line length, wave speed, component impedance and source timing.

Structure plan. Map the resulting pressure field into tank acoustic modes and then panel response. A common-volume pressure is not automatically spatially uniform at high frequency. Retain modal phase and pressure shape so that structural generalized forces are integrated rather than inferred from one microphone.

KSC-STD-164 Revision C notes that cross-spectral density magnitude/phase and coherence can help evaluate vibration transmission in large fixtures [5]. Its scope is Kennedy Space Center ground-support-equipment environmental testing, not a universal requirement for cryogenic feed systems. The signal-processing principle is transferable; the document's requirements are not.

05 / RETAIN AND APPLY

Three takeaways

1. RSS is an assumption. It follows from negligible cross terms, not from the fact that sources are physically separate.

2. Coherence alone is incomplete. The complex cross-spectrum phase decides whether correlated contributions reinforce or cancel.

3. Preserve phase through the whole chain. Source timing, feedline transfer, junction impedance, tank acoustic modes and panel response can all rotate phase.

Reflection

Four equal 1 psi RMS tones enter a common volume. What combined level should be used if only the individual autospectra are known?

Compact answer. There is no unique value. RSS gives 2 psi RMS (+6.02 dB) only under an incoherent assumption. Perfect in-phase addition gives 4 psi RMS (+12.04 dB), and coherent cancellation can be lower. Report a bounded sensitivity or obtain simultaneous cross-spectra; do not present RSS as measured truth.

Sources and calculation notes

[1] Holster, J. L., Astleford, W. J., and Gerlach, C. R. (May 1973). Analysis of Propellant Feedline Dynamics. NASA-CR-124308. Amplitude/phase transfer modeling with line and component effects.

[2] Michalopoulos, C. D., Clark, R. W., Jr., and Doiron, H. H. (1992). Acoustic Modes in Fluid Networks. Pressure and flow modes for branched one-dimensional networks.

[3] Hess, D. E. (December 1991). Spectral Analysis on a PC. NISTIR 4733. Definitions and estimation guidance for cross-spectral density, phase and coherence.

[4] Karchmer, A. M. (September 1977). Identification and Measurement of Combustion Noise from a Turbofan Engine Using Correlation and Coherence Techniques. NASA-TM-73747.

[5] NASA Kennedy Space Center (April 18, 2022). KSC-STD-164 Revision C: Environmental Test Methods for Ground Support Equipment. See Section 3.3.6.5.3; scope is KSC GSE testing.

Verified October 2, 2026. Figures and table: original illustrative calculations. Pressure magnitudes are RMS phasors at one common plane. No cryogenic-fluid property, hardware geometry or flight measurement is asserted.